Problem 1:

a) Since Mia has 3 EUR and she will spend all the money, she will be unpacking 3 eggs. Then, an outcome ω can be represented by $\omega = (\omega_1, \omega_2, \omega_3)$, where $\omega_i \in \{T, J\}$ for $i = 1, 2, 3$. and T denotes the unpacking a Tom figure and J a Jerry figure:

The probability space is given by:

i) The sample space $\Omega = \{T, J\}^3 = \{(\omega_1, \omega_2, \omega_3) \mid \omega_i \in \{T, J\} \text{ for } i = 1,2,3\}$

ii) The event space $\mathcal{F} = P(\Omega)$ since Ω is finite.

iii) The probability measure $\mathbb{P}(A) = \frac{|A|}{|\Omega|}$, where $A \in \mathcal{F}$ $(A \subseteq \Omega)$

Let B be the event of obtaining exactly one Tom figure, then
$\mathbb{P}(B) = \mathbb{P}(\{(T, J, J), (J, T, J), (J, J, T)\}) = \frac{|B|}{|\Omega|} = \frac{3}{8}$

Let C be the event of obtaining more Jerry than Tom figures, i.e., $C = \{(J, J, J)\} \cup B$, then $\mathbb{P}(C) = \frac{|C|}{|\Omega|} = \frac{4}{8} = \frac{1}{2}$

b) Since Mia stops buying surprise eggs until she has at least one Jerry figure, then the probability space $(\Omega, \mathcal{F}, \mathbb{P})$ is given by:

i) $\Omega = \{(J), (T, J), (T, T, J), (T, T, T)\}$ (sample space), with J and T defined as before

ii) $\mathcal{F} = P(\Omega)$ (Ω is finite)

iii) $\mathbb{P}(A) = \sum_{\omega \in A} p(\omega)$, where $A \in \mathcal{F}$ and $p(\omega) = \left(\frac{1}{2}\right)^{|\omega|}$, $|\omega|$:= length of tuple ω

Let B, C be events defined as before, then $\mathbb{P}(B) = \mathbb{P}(\{(T, J)\}) = \left(\frac{1}{2}\right)^2 = \frac{1}{4}$ and
$\mathbb{P}(C) = \mathbb{P}(\{(J)\}) = \left(\frac{1}{2}\right)^1 = \frac{1}{2}$

c) Since Mia stops buying eggs once she has at least one figure of each, the probability space $(\Omega, \mathcal{F}, \mathbb{P})$ is given by:

i) $\Omega = \{(J, J, J), (J, J, T), (J, T), (T, T, T), (T, T, J), (T, J)\}$, with J and T as before.

ii) $\mathcal{F} = P(\Omega)$ (as Ω is finite)

iii) $\mathbb{P}(A) = \sum_{\omega \in A} p(\omega)$, where $A \in \mathcal{F}$ and $p(\omega) = \left(\frac{1}{2}\right)^{|\omega|}$, $|\omega|$:= length of tuple ω

Let B, C be events defined as before, then $\mathbb{P}(B) = \mathbb{P}(\{(J, J, T), (J, T), (T, J)\})$
$= \left(\frac{1}{2}\right)^3 + \left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^2 = \frac{5}{8}$ and $\mathbb{P}(C) = \mathbb{P}(\{(J, J, J), (J, J, T)\}) = \left(\frac{1}{2}\right)^3 + \left(\frac{1}{2}\right)^3 = \frac{1}{4}$